

Provisional Product Data Sheet

SUMITOMO POLYETHYLENE

FS153S

Linear Low Density, Blown Film

Applications	For heavy duty liner, carrier bag, garment packaging, mulch film, and other general purpose film applications.
Characteristics	High strength film with excellent processability, high slip and antiblocking properties. For higher transparency, blending with LDPE is recommended.
Typical Processing Conditions	Melt Temperature 180 – 220°C Blow-up Ratio 2 – 3

Typical Properties

Physical		Value	Unit	Method
Melt Index	2.16kg	1	g/10min	ASTM D1238
Density		0.923	g/cm ³	ASTM D792-A
Thermal		Value	Unit	Method
Vicat Softening Temperature	10N	100	°C	ASTM D1525A
Melt Temperature		121	°C	DSC
Crystallisation Temperature		108	°C	DSC
Brittleness Temperature		< -50	°C	ASTM D746
Film		Value	Unit	Method
Tensile Strength at Break	MD / TD	38 / 31	MPa	ASTM D882
Tensile Elongation at Break	MD / TD	690 / 750	%	ASTM D882
Tensile Modulus, 1% Secant	MD / TD	215 / 260	MPa	ASTM D882
Elmendorf Tear Strength	MD / TD	160 / 345	g	ASTM D1922
Blocking Load		11	g/100cm ²	ASTM D3354*
Coefficient of Friction, Exterior	μ_s / μ_k	0.12 / 0.09		ASTM D1894
Dart Drop Impact		125	g	ASTM D1709
Haze		16	%	ASTM D1003
Gloss	45°	50	%	ASTM D2457

* modified

The values shown are typical values measured on the product and are not to be considered as guaranteed specifications. Actual test results may vary.

Test pieces preparation condition as per ASTM D 4976 and ISO 1872-2.

Blown film extruder ϕ 55mm, die ϕ 125mm, die gap 2.0mm, extrusion temperature 190°C, output 25kg/hr, BUR 2.0, thickness 30 μ m.

Food Contact Application

Unmodified FS153S complies with the US FDA Regulation 21 CFR 177.1520 (c) 3.2a for use in direct contact with food as well as with the EU Directive 2002/72/EC and its amendments 2004/1/EC, 2004/19/EC, 2005/79/EC, and 2007/19/EC.

Restriction of Hazardous Substances

The substances regulated under EU Directive 2002/95/EC on RoHS in electrical and electronic equipment are not used in the manufacturing process of FS153S and are therefore not expected to be present.

Storage and Handling

FS153S is supplied in 25kg bags or in bulk containers depending on packaging options available for different markets. It is advisable to store in a well ventilated storage facility, protected from rain, direct exposure to sunlight, temperature, and dust.

Moisture may accumulate inside the packaging if there is a significant fluctuation in ambient temperature and high relative humidity. If this occurs, customer is strongly advised to dry the resin prior to use. Direct exposure to sunlight and excessive ambient temperature may affect the properties, appearance, and odour characteristics of the resin.

Please refer to the Material Safety Data Sheet (MSDS) for information regarding industrial health and safety issues.

Disposal and the Environment

SCA prioritizes environmental management in all our operations in responsibility and respect towards the community and the world around us.

Unmodified, our polypropylene and polyethylene products are readily recyclable, both for reuse via reprocessing, and for energy reclamation through incineration. If disposed in landfills, our products, which are inert materials, do not pose pollution risks since they do not generate gaseous or liquid emissions.

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